

Chapter

5

Normal Probability Distributions

GLOBAL
EDITION



Elementary Statistics

Picturing the World

EIGHTH EDITION

Ron Larson



Chapter Outline

- 5.1 Introduction to Normal Distributions and the Standard Normal Distribution
- 5.2 Normal Distributions: Finding Probabilities
- 5.3 Normal Distributions: Finding Values
- 5.4 Sampling Distributions and the Central Limit Theorem
- 5.5 Normal Approximations to Binomial Distributions

Section 5.1

Introduction to Normal Distributions and the Standard Normal Distribution

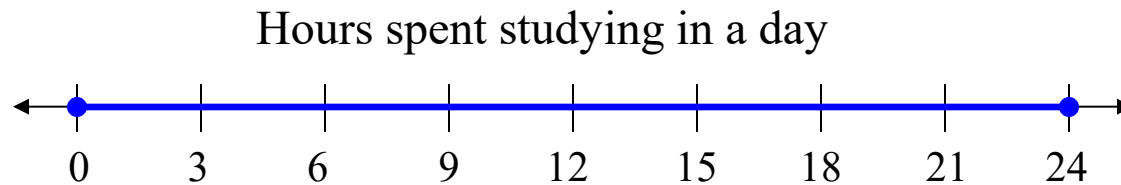
Section 5.1 Objectives

- How to interpret graphs of normal probability distributions
- How to find areas under the standard normal curve

Properties of a Normal Distribution

Continuous random variable

- Has an infinite number of possible values that can be represented by an interval on the number line.



The time spent studying can be any number between 0 and 24.

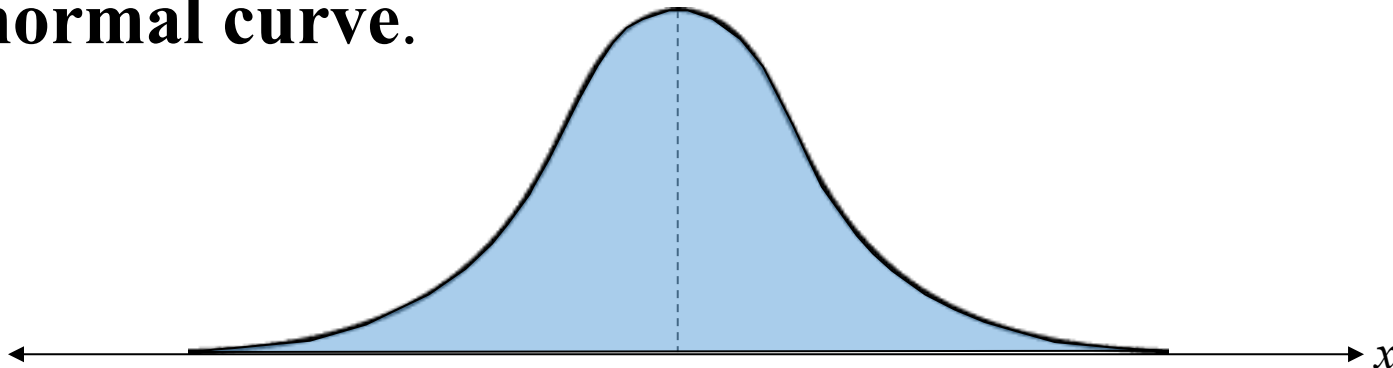
Continuous probability distribution

- The probability distribution of a continuous random variable.

Properties of a Normal Distribution

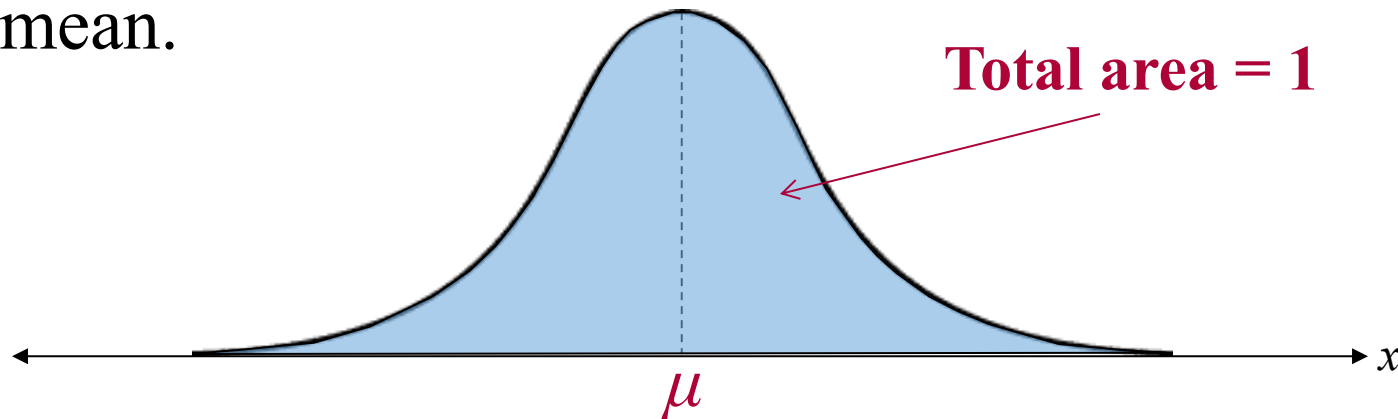
Normal distribution

- A continuous probability distribution for a random variable, x .
- The most important continuous probability distribution in statistics.
- The graph of a normal distribution is called the **normal curve**.



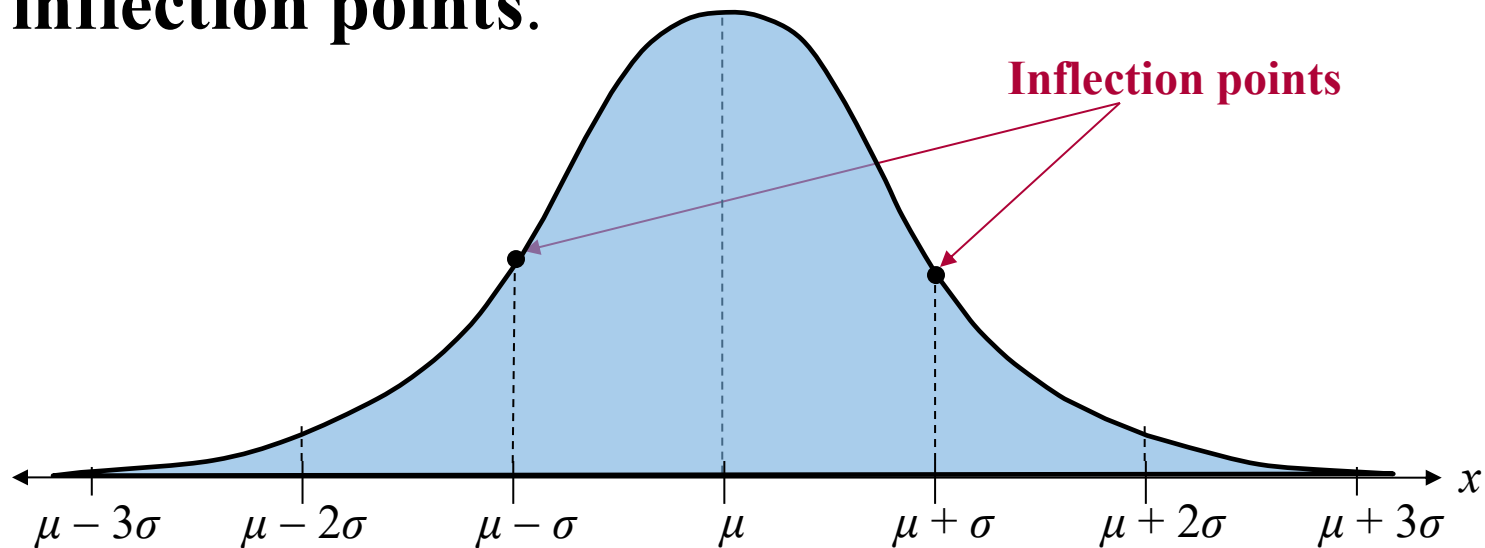
Properties of a Normal Distribution

1. The mean, median, and mode are equal.
2. The normal curve is bell-shaped and is symmetric about the mean.
3. The total area under the normal curve is equal to one.
4. The normal curve approaches, but never touches the x -axis as it extends farther and farther away from the mean.



Properties of a Normal Distribution

5. Between $\mu - \sigma$ and $\mu + \sigma$ (in the center of the curve), the graph curves downward. The graph curves upward to the left of $\mu - \sigma$ and to the right of $\mu + \sigma$. The points at which the curve changes from curving upward to curving downward are called the **inflection points**.

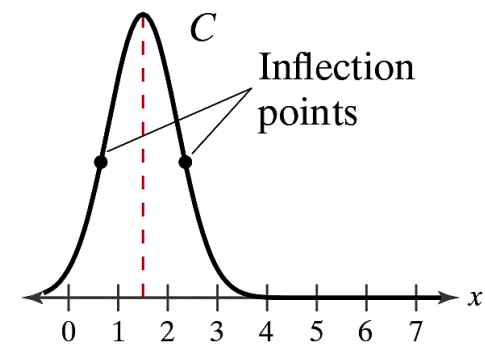
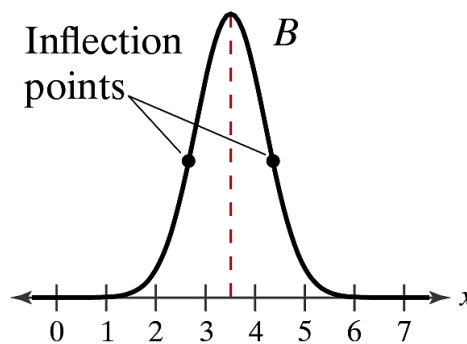
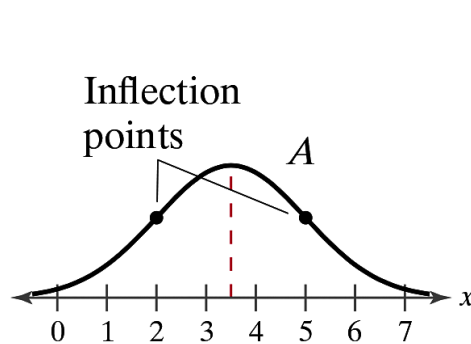


Probability Density Function (PDF)

- A discrete probability distribution can be graphed with a histogram.
- For a continuous probability distribution, you can use a **probability density function (pdf)**.
- A probability density function has two requirements:
 1. the total area under the curve is equal to 1
 2. the function can never be negative.

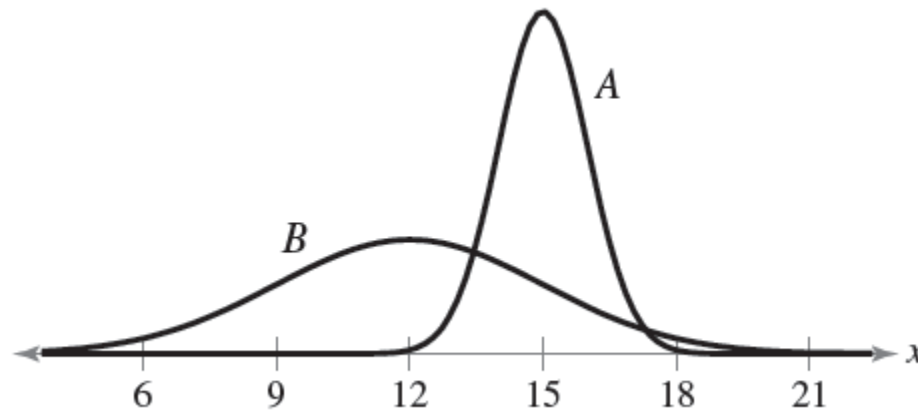
Means and Standard Deviations

- A normal distribution can have any mean and any positive standard deviation.
- The mean gives the location of the line of symmetry.
- The standard deviation describes the spread of the data.



Example: Understanding Mean and Standard Deviation

1. Which curve has the greater mean?

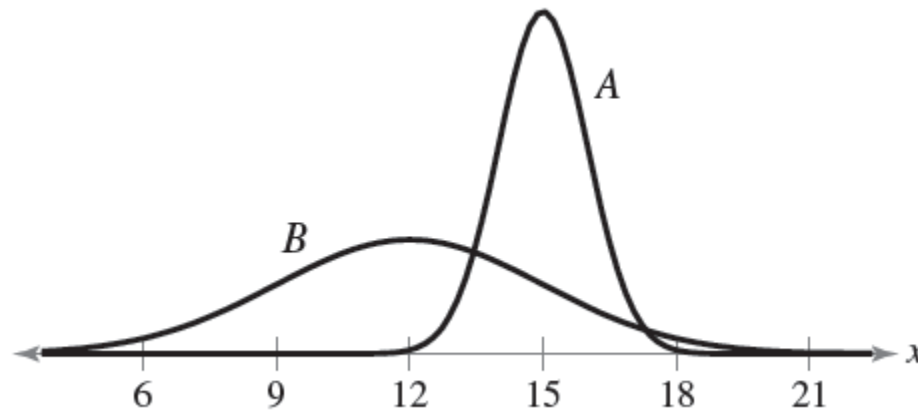


Solution:

Curve A has the greater mean (The line of symmetry of curve A occurs at $x = 15$. The line of symmetry of curve B occurs at $x = 12$.)

Example: Understanding Mean and Standard Deviation

2. Which curve has the greater standard deviation?

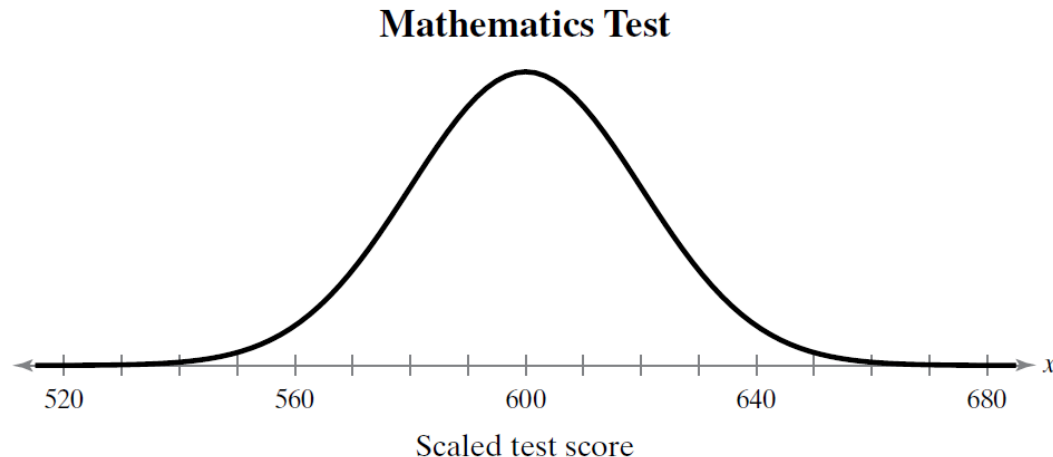


Solution:

Curve *B* has the greater standard deviation (Curve *B* is more spread out than curve *A*.)

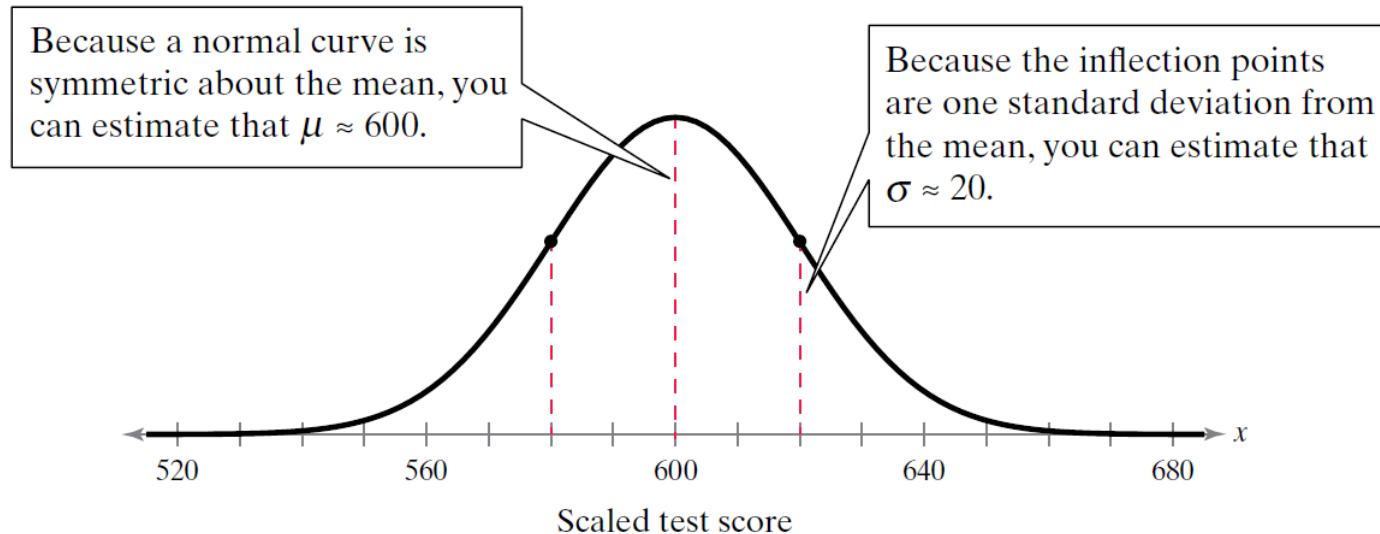
Example: Interpreting Graphs of Normal Distributions

The scaled test scores for New York State Grade 6 Common Core Mathematics Test are normally distributed. The normal curve shown below represents this distribution. What is the mean test score? Estimate the standard deviation of this normal distribution.
(Adapted from New York State Education Department)



Solution: Interpreting Graphs of Normal Distributions

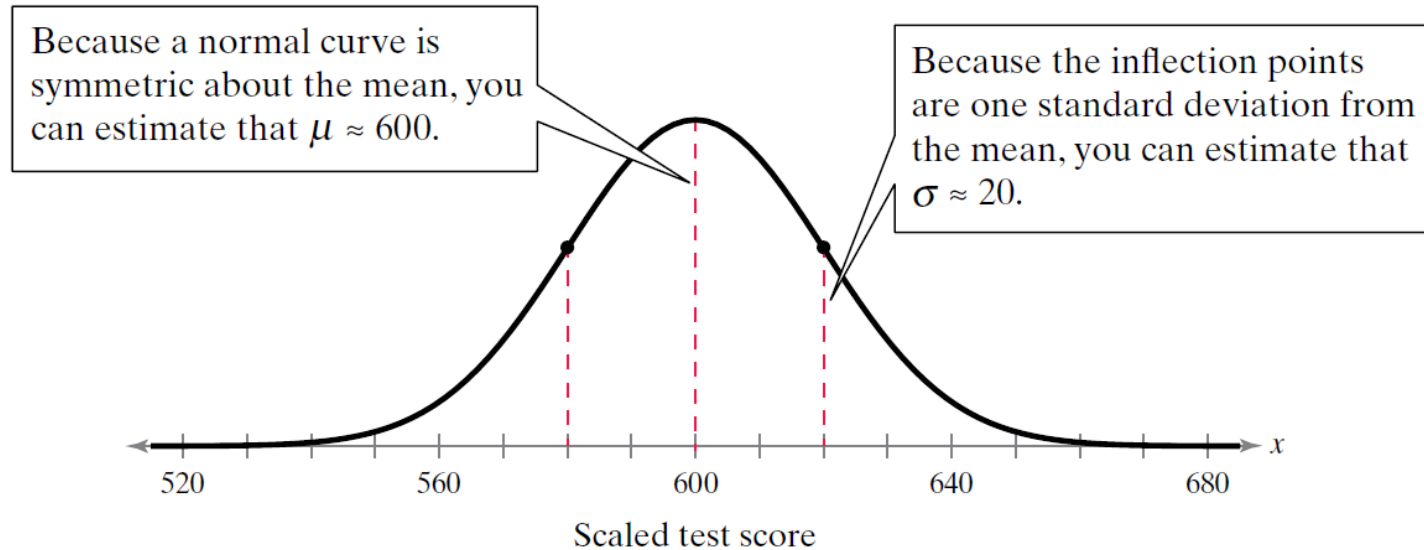
Solution:



The scaled test scores for the New York State Grade 6 Common Core Mathematics Test are normally distributed with a mean of about 600 and a standard deviation of about 20.

Solution: Interpreting Graphs of Normal Distributions

Solution:

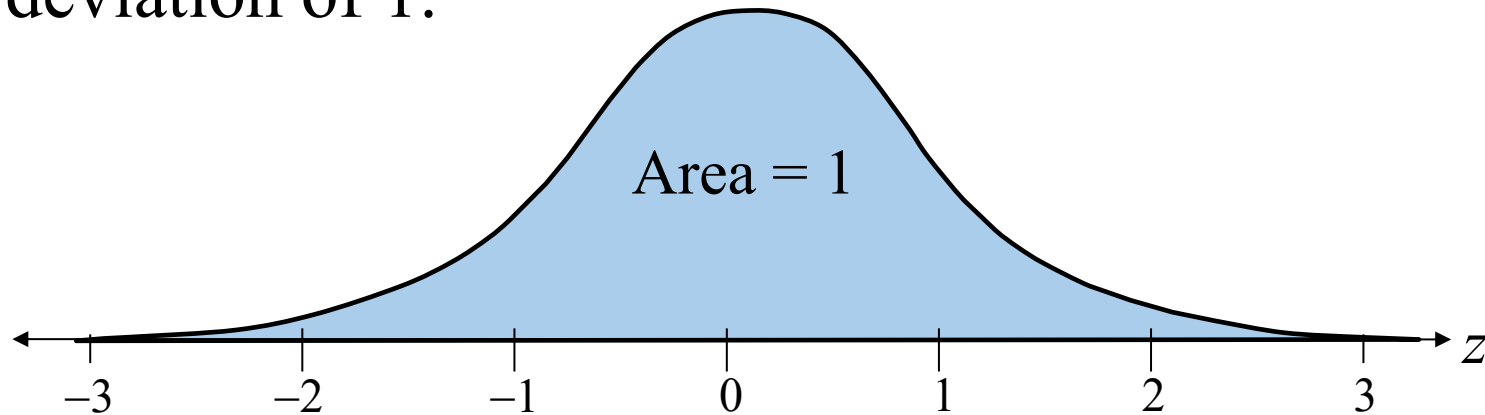


Using the Empirical Rule, you know that about 68% of the scores are between 580 and 620, about 95% of the scores are between 560 and 640, and about 99.7% of the scores are between 540 and 660.

The Standard Normal Distribution

Standard normal distribution

- A normal distribution with a mean of 0 and a standard deviation of 1.

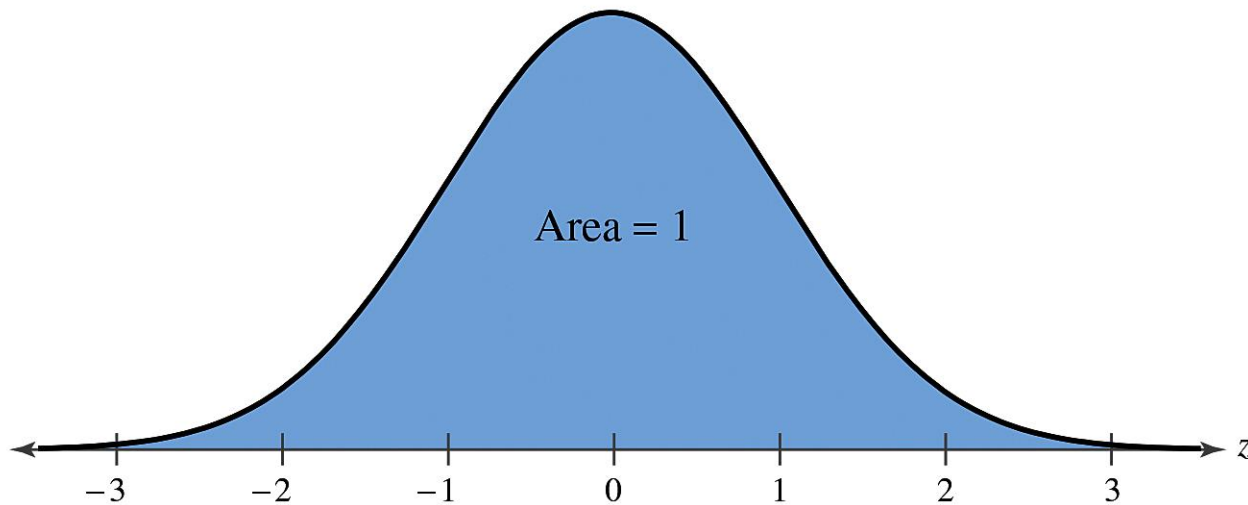


- Any x -value can be transformed into a z -score by using the formula

$$z = \frac{\text{Value} - \text{Mean}}{\text{Standard deviation}} = \frac{x - \mu}{\sigma}$$

The Standard Normal Distribution

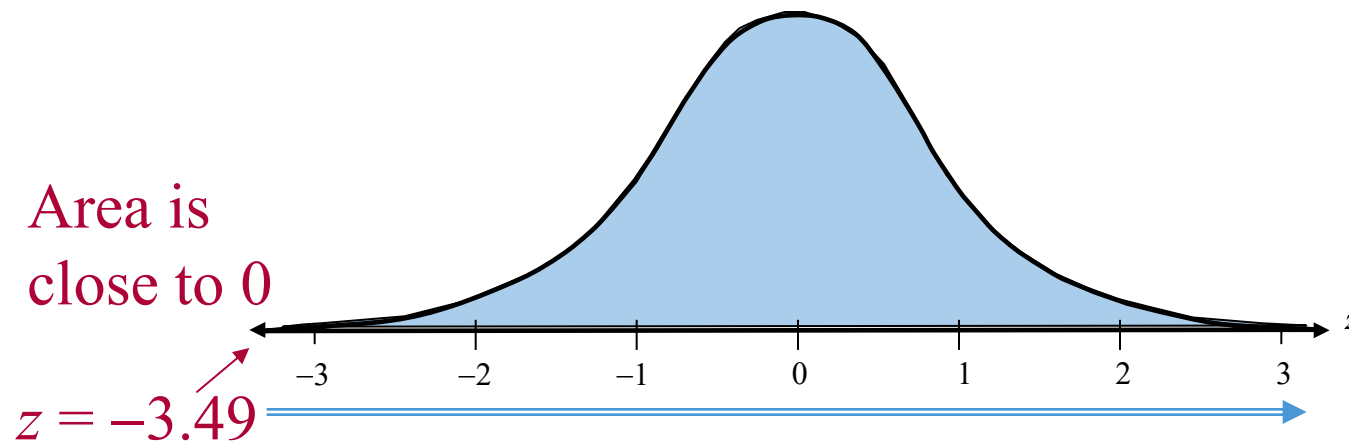
The **standard normal distribution** is a normal distribution with a mean of 0 and a standard deviation of 1. The total area under its normal curve is 1.



Standard Normal Distribution

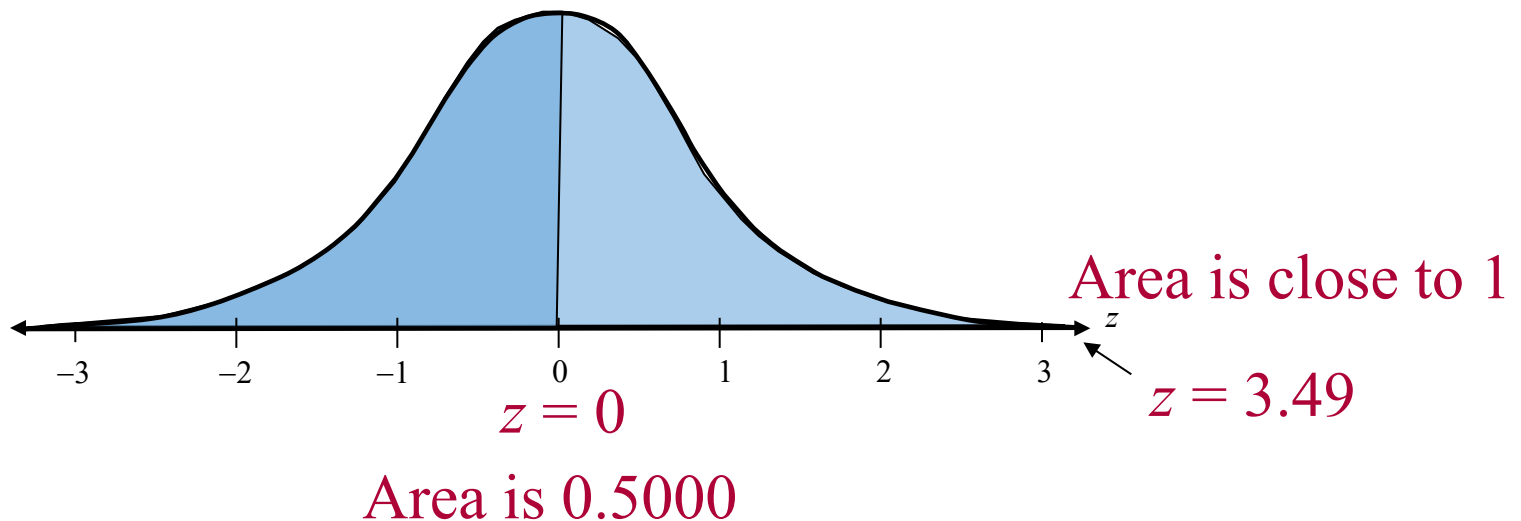
Properties of the Standard Normal Distribution

1. The cumulative area is close to 0 for z -scores close to $z = -3.49$.
2. The cumulative area increases as the z -scores increase.



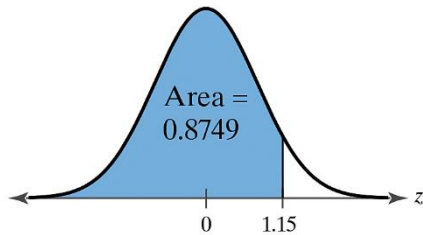
Properties of the Standard Normal Distribution

3. The cumulative area for $z = 0$ is 0.5000.
4. The cumulative area is close to 1 for z -scores close to $z = 3.49$.



Example: Using The Standard Normal Table

- Find the cumulative area that corresponds to a z-score of 1.15.



z	.00	.01	.02	.03	.04	.05	.06
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026

0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279

Solution:

Find 1.1 in the left hand column.

Move across the row to the column under 0.05.

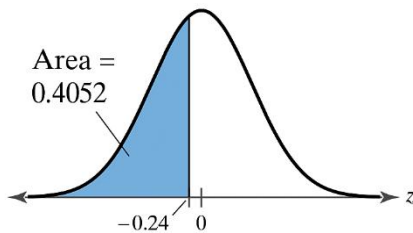
The area to the left of $z = 1.15$ is 0.8749.

TI-84 PLUS

```
normalcdf(-10000
,1.15,0,1)
.8749280114
```

Example: Using The Standard Normal Table

2. Find the cumulative area that corresponds to a z-score of -0.24 .



z	.09	.08	.07	.06	.05	.04	.03
-3.4	.0002	.0003	.0003	.0003	.0003	.0003	.0003
-3.3	.0003	.0004	.0004	.0004	.0004	.0004	.0004
-3.2	.0005	.0005	.0005	.0006	.0006	.0006	.0006
-0.5	.2776	.2810	.2843	.2877	.2912	.2946	.2981
-0.4	.3121	.3156	.3192	.3228	.3264	.3300	.3336
-0.3	.3483	.3520	.3557	.3594	.3632	.3669	.3707
-0.2	.3859	.3897	.3936	.3974	.4013	.4052	.4090
-0.1	.4247	.4286	.4325	.4364	.4404	.4443	.4483
-0.0	.4641	.4681	.4721	.4761	.4801	.4840	.4880

Solution:

Find -0.2 in the left hand column.

Move across the row to the column under 0.04 .

The area to the left of $z = -0.24$ is 0.4052 .

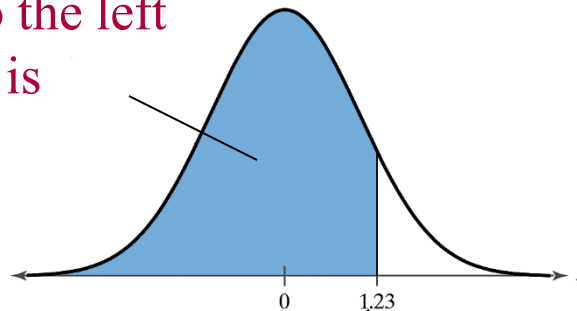
TI-84 PLUS

```
normalcdf(-10000,
, -.24, 0, 1)
.405165175
```

Finding Areas Under the Standard Normal Curve

1. Sketch the standard normal curve and shade the appropriate area under the curve.
2. Find the area by following the directions for each case shown.
 - a. To find the area to the *left* of z , find the area that corresponds to z in the Standard Normal Table.

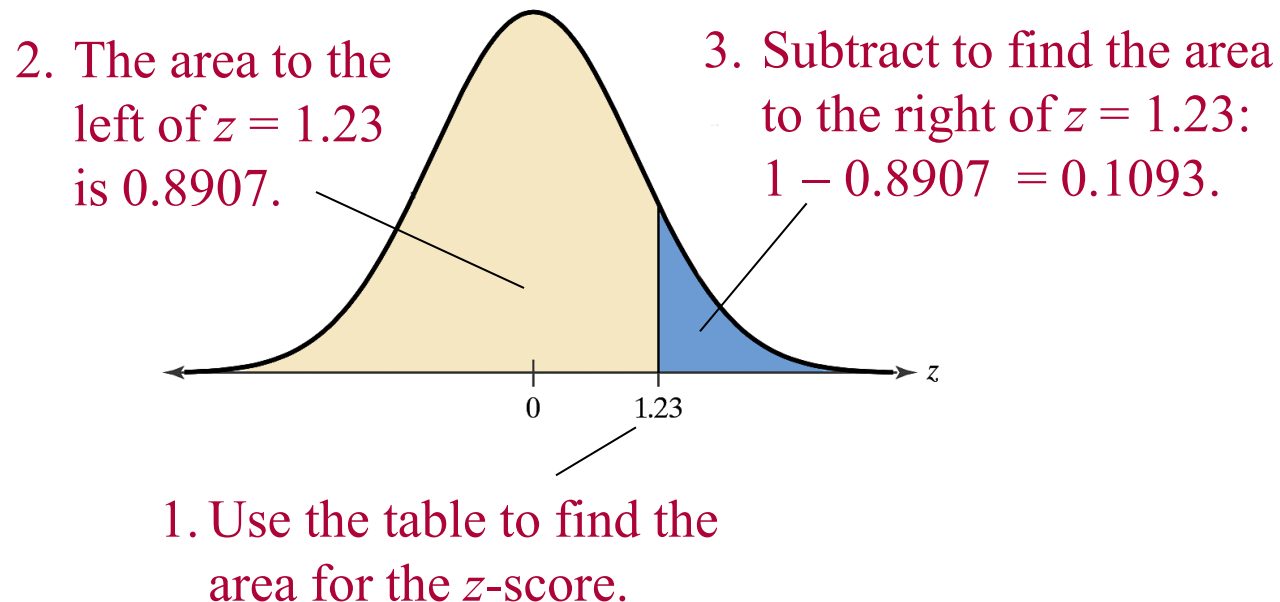
2. The area to the left of $z = 1.23$ is 0.8907.



1. Use the table to find the area for the z -score.

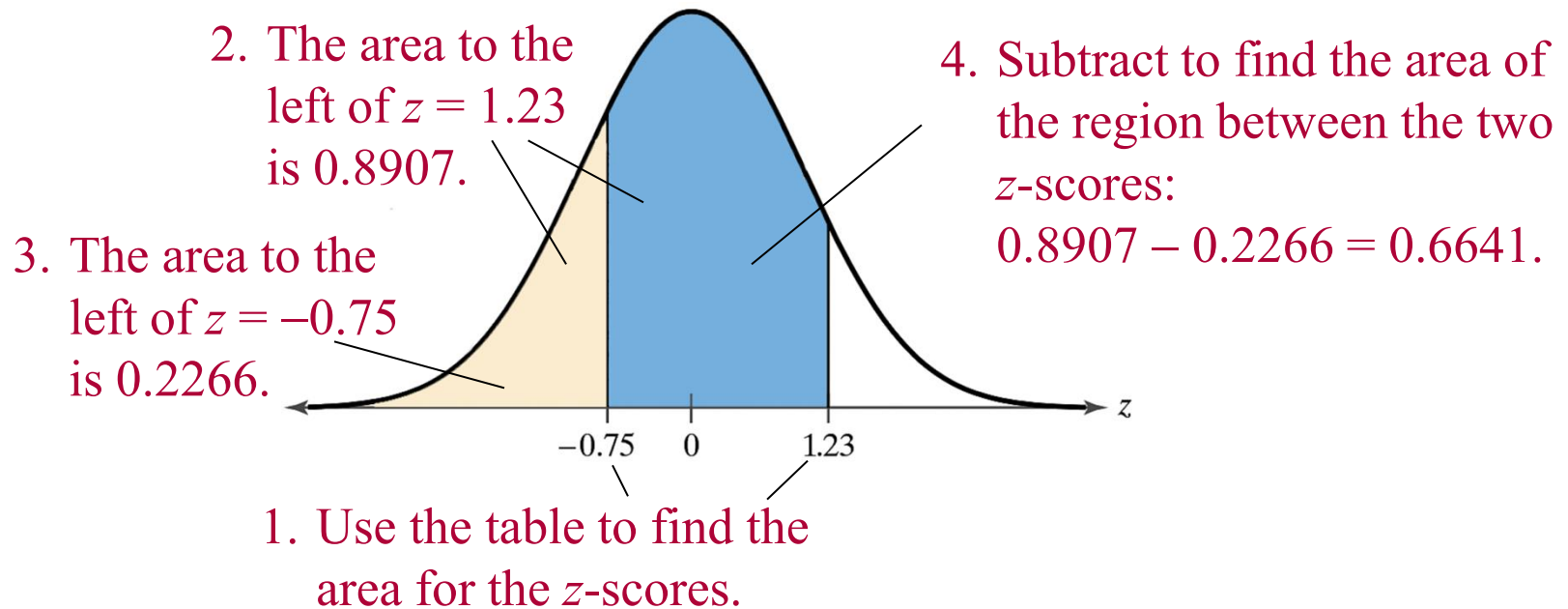
Finding Areas Under the Standard Normal Curve

- b. To find the area to the *right* of z , use the Standard Normal Table to find the area that corresponds to z . Then subtract the area from 1.



Finding Areas Under the Standard Normal Curve

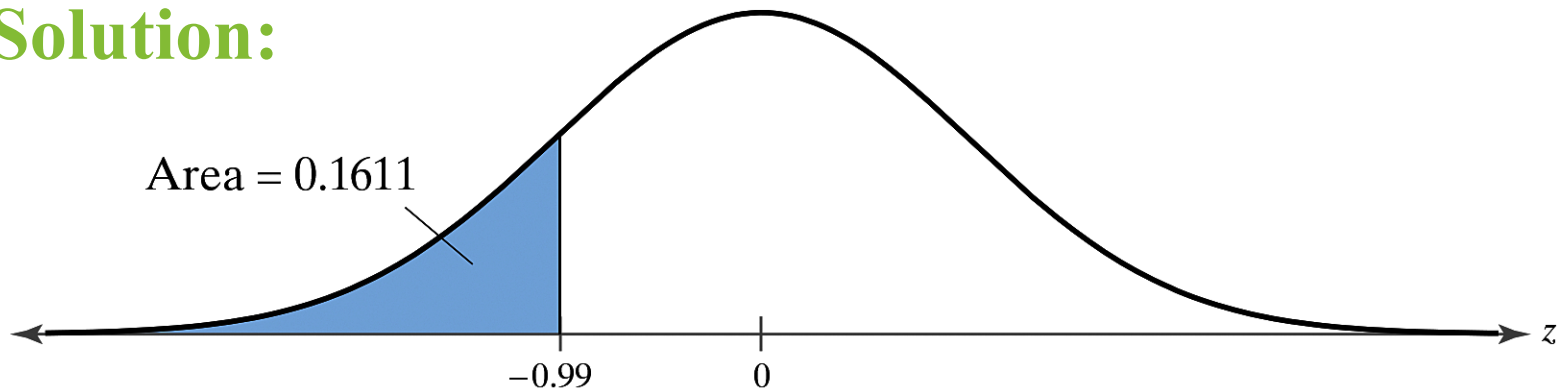
- c. To find the area *between* two z-scores, find the area corresponding to each z-score in the Standard Normal Table. Then subtract the smaller area from the larger area.



Example: Finding Area Under the Standard Normal Curve

Find the area under the standard normal curve to the left of $z = -0.99$.

Solution:



From the Standard Normal Table, the area is equal to 0.1611.

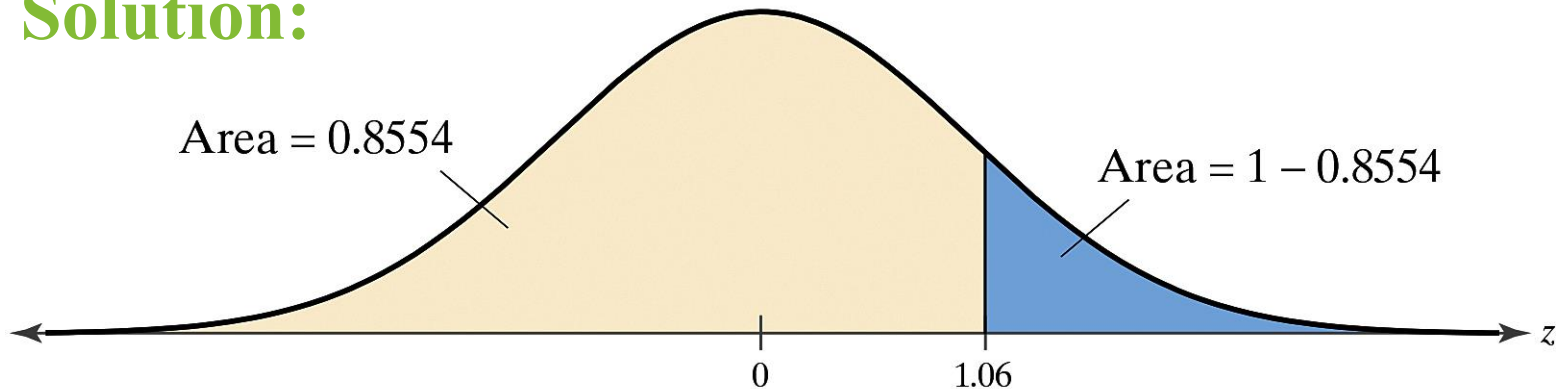
EXCEL

	A	B
1	NORM.S.DIST(-0.99,TRUE)	
2		0.16108706

Example: Finding Area Under the Standard Normal Curve

Find the area under the standard normal curve to the right of $z = 1.06$.

Solution:



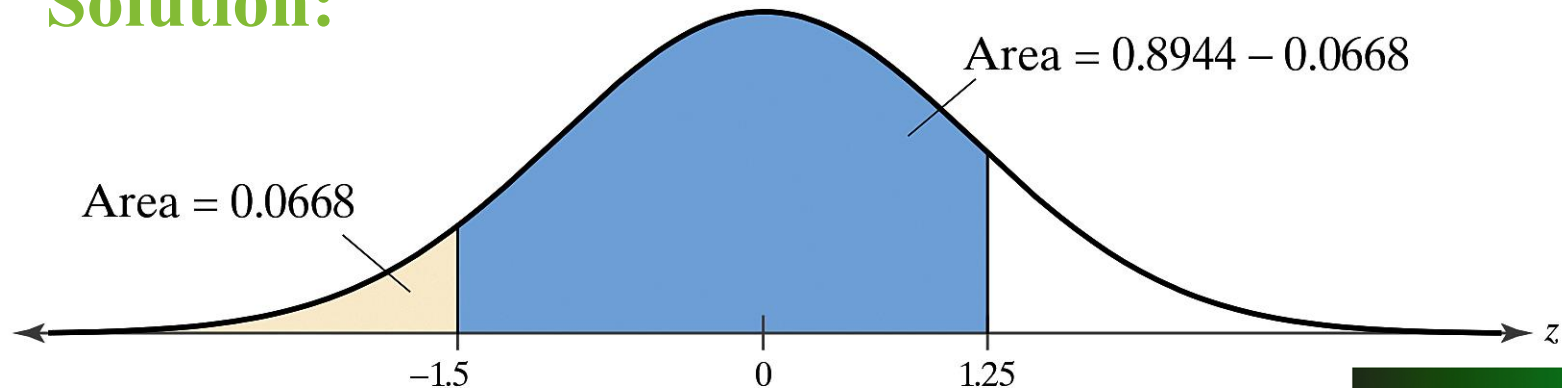
From the Standard Normal Table, the area is equal to 0.1446.

EXCEL		
	A	B
1	1-NORM.S.DIST(1.06,TRUE)	
2		0.1445723

Example: Finding Area Under the Standard Normal Curve

Find the area under the standard normal curve between $z = -1.5$ and $z = 1.25$.

Solution:



From the Standard Normal Table, the area is equal to 0.8276. So, 82.76% of the area under the curve falls between $z = -1.5$ and $z = 1.25$.

TI-84 PLUS

```
normalcdf(-1.5,1.25,0,1)  
.8275429323
```